

Analysis of Cryptocurrency Transaction Characteristics and Trend Prediction Based on Machine Learning: A Case Study of BTC and ETH Data from March to May 2025

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Abstract

With the maturation of blockchain technology and the rapid development of the cryptocurrency market, Bitcoin (BTC) and Ethereum (ETH) have become the most influential digital assets globally. This study, based on a total of 50,000 real transaction data from March to May 2025, systematically analyzed the transaction characteristics, user behavior patterns, and market trends of BTC and ETH. Through methods such as data cleaning, exploratory analysis, statistical modeling, and machine learning, it revealed the similarities and differences between the two cryptocurrencies in terms of transaction volume, transaction costs, and time distribution, and constructed models for transaction currency classification and transaction fee prediction. The research found that BTC and ETH have significant differences in transaction activity, fee structure, and user behavior, and the market shows short-term volatility and structural features. This study provides constructive policy recommendations for participants, investors, and regulatory authorities in the cryptocurrency market.

Key words: Cryptocurrency ; Bitcoin ; Ethereum ; Trading behavior ; Machine learning

I. Introduction

1. Research Background

In recent years, cryptocurrencies have evolved from a marginal technological innovation to a significant component of the global financial system. Bitcoin, as the first decentralized digital currency, is renowned for its scarcity and security; Ethereum, on the other hand, has expanded the application scenarios of blockchain through its smart contract functionality. The differences in trading behaviors, market dynamics, and user groups between the two have become a hot topic of interest in both academic and industrial circles. With the integration of data science and blockchain technology, behavioral analysis based on real transaction data offers a new perspective for understanding market mechanisms.

2. Research Objectives and Significance

This study aims to conduct an in-depth analysis of the trading data of BTC and ETH to

understand the similarities and differences in their basic characteristics such as trading volume, trading amount, and transaction fees, explore the patterns of user address trading behavior, compare the behavioral differences between high-frequency traders and ordinary users, and investigate how to predict the type of traded currency and transaction fees through machine learning models and predict the short-term market trend based on the current trend.

The theoretical and practical contributions based on the research results include: First, analyzing trading behavior characteristics and market trends to assist investors and traders in making investment decisions, and providing data support for regulatory authorities to implement supervision measures. Second, revealing the activity patterns of cryptocurrency users for product design, risk control, and marketing. Third, demonstrating the application of data science methods in blockchain data analysis and providing references for subsequent research.

II. Research Contents

1 Data

1.1 Data Source and Overview

The data for this study was sourced from the Kaggle dataset platform (<https://www.kaggle.com/>). It comprises 50,000 cryptocurrency transaction records covering trading information for Bitcoin (BTC) and Ethereum (ETH) from March to May 2025. The data fields include Transaction ID, Sender Address, Receiver Address, Transaction Amount, Transaction Fee, Timestamp, Block ID, Mining Pool, Currency Type, Transaction Type, Transaction Status, and Gas Price (exclusive to ETH).

1.2 Data Cleaning and Preprocessing

1.2.1 Missing Value Handling

The 'Gas Price' field is exclusive to ETH transactions; missing values for BTC transactions were imputed as 0. There were no missing values in other fields.

1.2.2 Timestamp Parsing

Timestamps were converted into dimensions such as date, hour, and day of the week to facilitate time series analysis.

1.2.3 Duplicate Detection

No duplicate records were found in the dataset, ensuring data uniqueness.

1.2.4 Data Type Conversion

Categorical variables were label-encoded, and numerical variables were standardized to adapt to model inputs.

After preprocessing, the dataset contained 50,000 valid records, consisting of 25,012 BTC transactions and 24,988 ETH transactions, indicating a roughly balanced distribution between the two currency types.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 12 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Transaction_ID       50000 non-null  object
1   Sender_Address       50000 non-null  object
2   Receiver_Address     50000 non-null  object
3   Amount               50000 non-null  float64
4   Transaction_Fee      50000 non-null  float64
5   Timestamp            50000 non-null  object
6   Block_ID             50000 non-null  object
7   Mining_Pool          50000 non-null  object
8   Currency             50000 non-null  object
9   Transaction_Type     50000 non-null  object
10  Transaction_Status    50000 non-null  object
11  Gas_Price_Gwei       25012 non-null  float64
dtypes: float64(3), object(9)
memory usage: 4.6+ MB
```

2 Analysis & Application

2.1 Exploratory Data Analysis

2.1.1 Data Overview

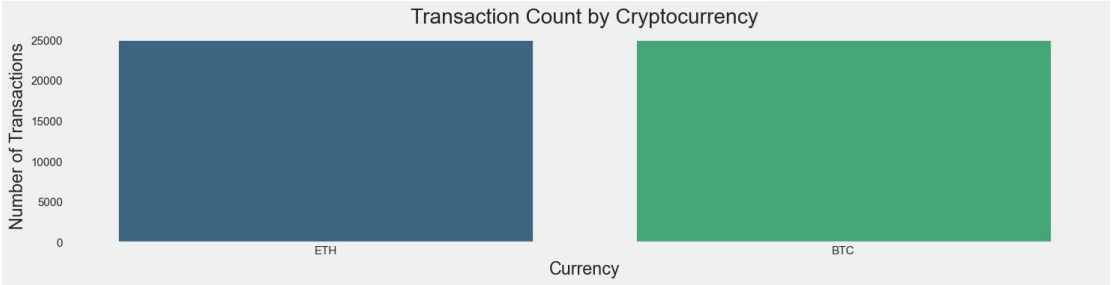


Fig. 1 Comparison of Transaction Volume between Bitcoin (BTC) and Ethereum (ETH)

After preprocessing, the dataset contains 50,000 valid records, with 25,012 BTC transactions and 24,988 ETH transactions. This indicates that the dataset is balanced regarding currency types, and no severe class imbalance issues exist.

Table 1 Descriptive Statistical Analysis

	Amount	Transaction_Fee	Gas_Price_Gwei	Hour	Day	Month	Year
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.0
mean	7.519087	0.000382	13.580300	11.419520	14.64870	3.738860	2025.0
std	4.321356	0.000257	16.403516	6.921722	8.77371	0.707471	0.0
min	0.006032	0.000010	0.000000	0.000000	1.00000	3.000000	2025.0
25%	3.784158	0.000177	0.000000	5.000000	7.00000	3.000000	2025.0
50%	7.514979	0.000340	10.000000	11.000000	14.00000	4.000000	2025.0
75%	11.234608	0.000506	25.000000	17.000000	22.00000	4.000000	2025.0
max	14.999943	0.001000	50.000000	23.000000	31.00000	5.000000	2025.0

The study focuses on data from March to May 2025, suitable for analyzing short-term trading behaviors. First, the transaction amount has a mean of 7.5, a median of 7.51, and a standard deviation of 4.32, indicating an approximately normal distribution with significant volatility. Second, the mean and median of the Transaction Fee are close, with a very small standard deviation, suggesting a highly symmetric and stable fee distribution. Finally, Gas_Price_Gwei exhibits a right-skewed distribution with a large standard deviation (16.40); the 25th percentile is 0 (BTC transactions), and the 50th percentile is 10 (ETH transactions), indicating a bimodal characteristic driven by different currency types.

The volatility in transaction amounts reflects diverse user purposes (micropayments, large transfers, test transactions). The stability of transaction fees stems partly from the high proportion of BTC (no Gas fees) and potentially from low congestion in the ETH network during this period, suggesting a light network load. The bimodal characteristic of Gas prices intuitively reflects the mechanism difference: BTC's 'no Gas' model versus ETH's 'Gas pricing' model leads to a split distribution.

2.1.2 Transaction Trends Over Time

Transaction trend analysis reveals dynamic changes in trading activity through time series, reflecting fluctuations in market or network status.

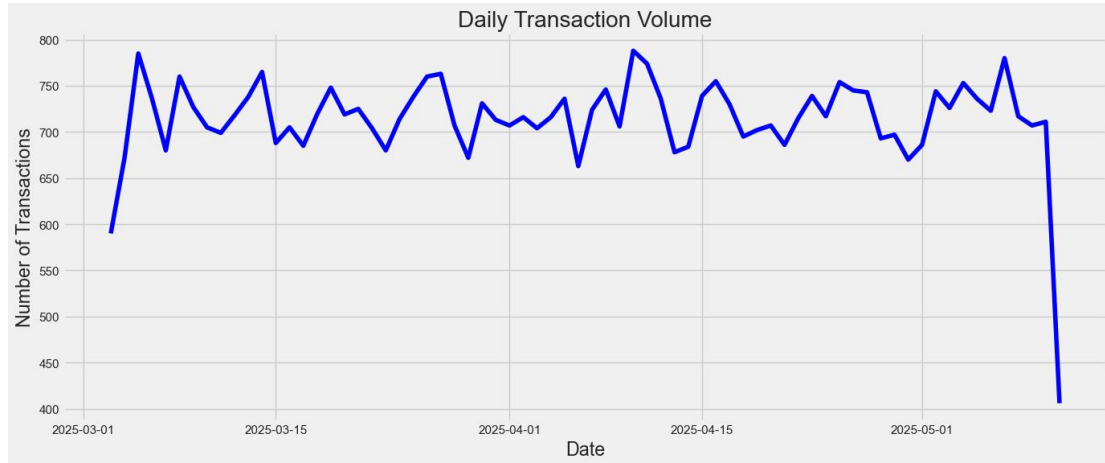


Fig. 2 Daily Total Transaction Volume

Between March and May 2025, daily transaction volumes fluctuated between 600 and 800, showing an overall stable trend without sustained growth or decline. However, there were obvious ‘jagged’ fluctuations; for instance, a peak of nearly 800 transactions occurred in early March, followed by a fluctuate decline, another peak in mid-April, and a sharp drop in early May due to missing data statistics..

The significant daily volatility may result from short-term market sentiment, network congestion, or external regulatory policies. However, users did not significantly alter their trading behaviors due to market volatility, maintaining the continuity of ‘daily transactions’. This stability is an embodiment of the blockchain ecosystem’s ‘daily operation’, reflecting its robustness as a decentralized payment system.

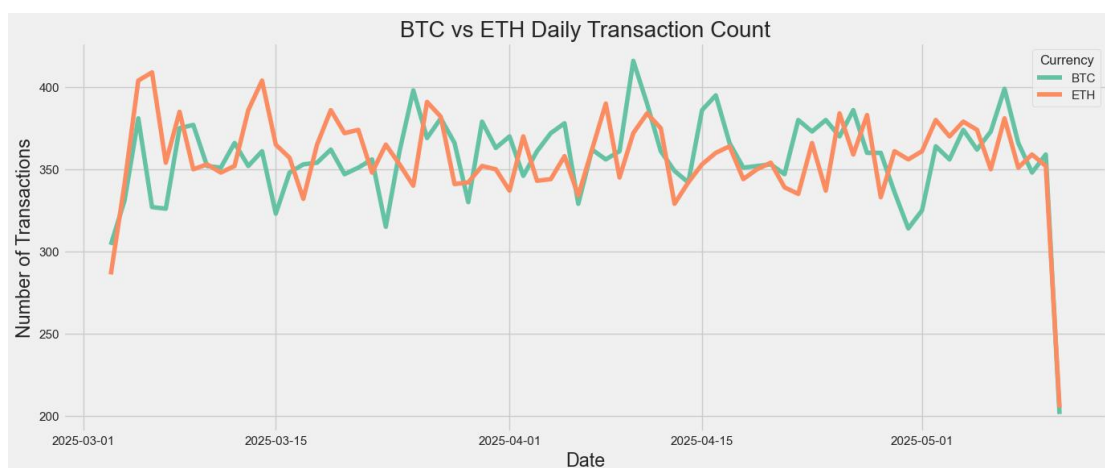


Fig. 3 Daily Transaction Trends for Bitcoin and Ethereum

Specifically, looking at BTC and ETH, the daily volume trends show a certain correlation. Locally, ETH volume exceeded BTC in mid-March and mid-April, while BTC was slightly higher

in late March and early May. This ‘alternating lead’ suggests different driving factors: ETH volume may be driven by applications like DeFi and NFTs, while BTC volume may be driven by value storage and cross-border payment needs. However, a favorable overall market environment impacts both.

2.1.3 Analysis of Transaction Amount and Transaction Fees

Analyzing transaction amounts and fees is central to EDA, revealing transaction scales and cost structures through distribution characteristics and correlations.

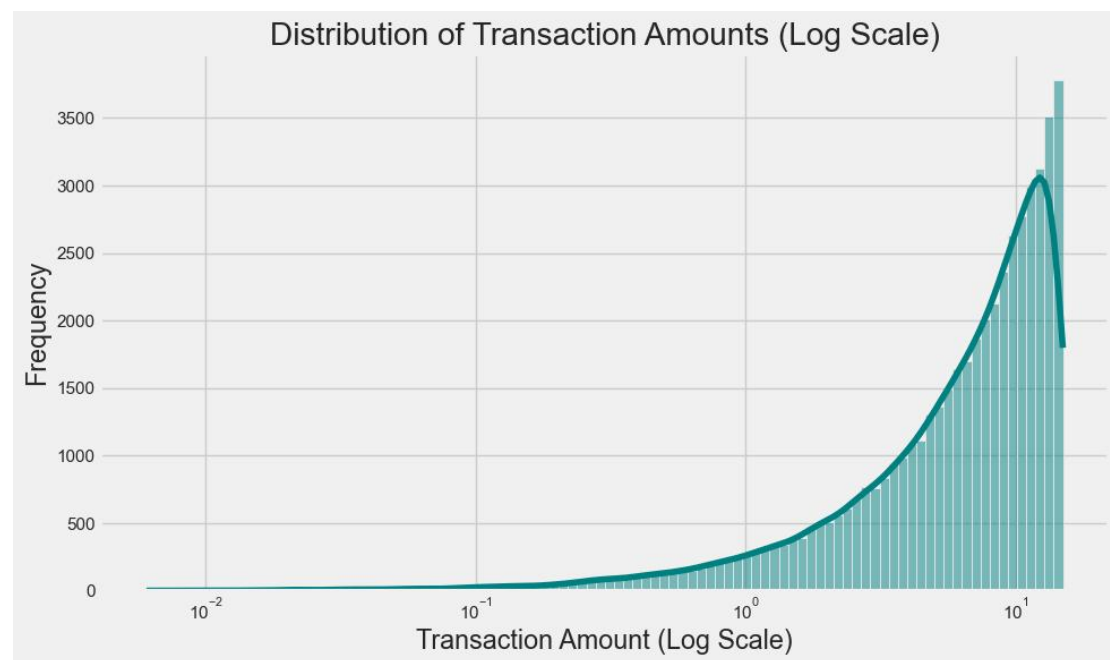


Fig. 4 Total Transaction Amount (Log Scale)

To analyze distribution characteristics, transaction amounts were visualized on a logarithmic scale. Overall, the distribution is right-skewed, with the peak concentrated in the small-to-medium range on the right side of the logarithmic scale. The frequency is high on the left and low on the right, presenting a ‘long-tail’ characteristic. The right-skewed distribution indicates a typical feature of cryptocurrency trading: a high proportion of small-value transactions and a low proportion of large-value transactions. The long tail suggests the existence of ‘whales’ or institutional users, though their proportion is minimal.

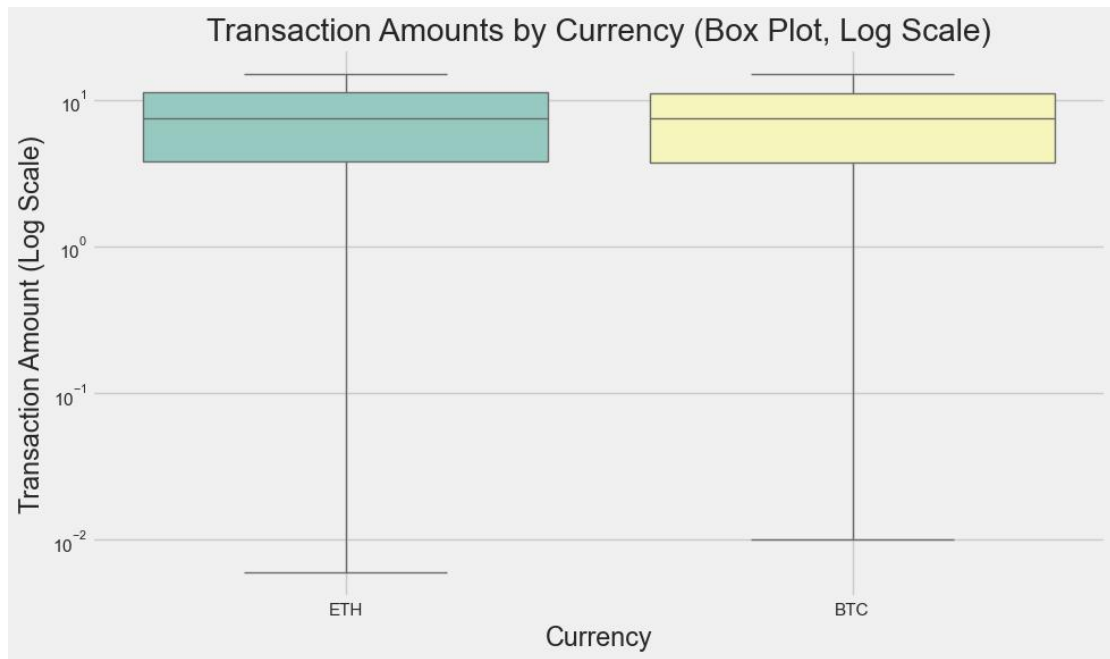


Fig. 5 Boxplot Comparison of Transaction Amounts for Bitcoin and Ethereum

Specifically, the medians for BTC and ETH are very close, indicating similar typical transaction sizes and similar user habits for daily transactions. However, the interquartile range (IQR) for ETH is slightly wider than for BTC, implying greater dispersion in medium-scale ETH transaction amounts. This may be due to the complexity of the ETH ecosystem (e.g., DeFi transactions involving multiple transfers and smart contract interactions). Conversely, BTC has a lower minimum transaction amount, indicating more extremely small transactions, consistent with its use cases in ‘value storage’ or ‘micropayments’.

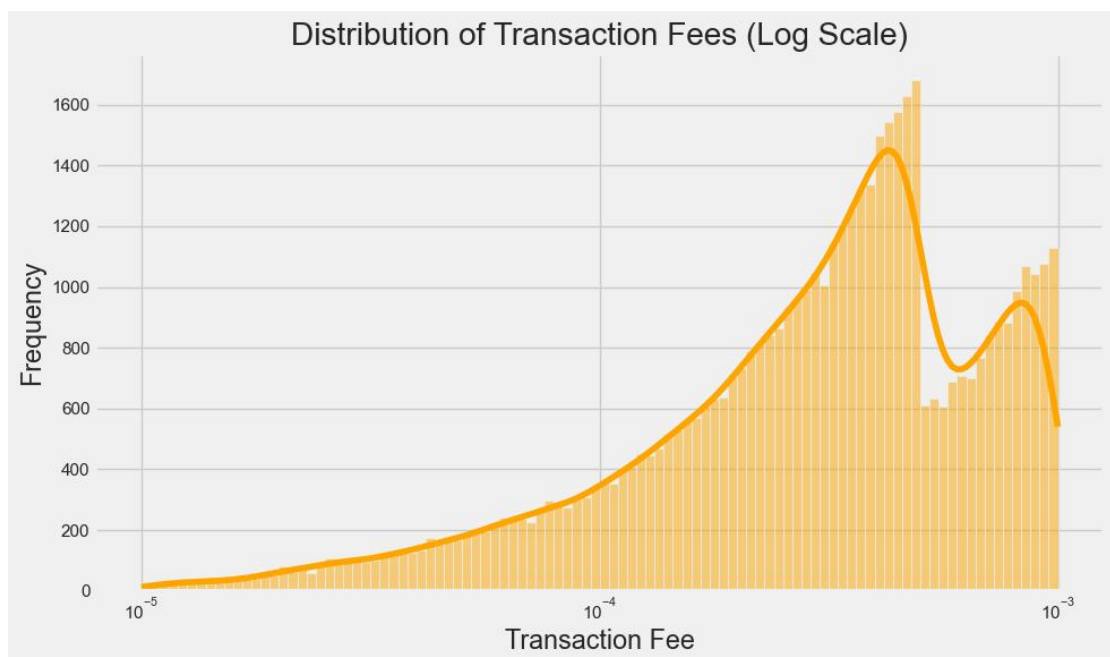


Fig. 6 Transaction Fee Distribution (Log Scale)

Besides transaction amount, transaction fees are a key indicator of trading characteristics. The distribution is visualized on a logarithmic scale. First, the KDE curve shows a clear bimodal pattern: the first peak (low fees, approx. 10^{-4} magnitude) corresponds to BTC transactions (no Gas fees), and the second peak (medium fees, approx. 10^{-3} magnitude) corresponds to ETH transactions (Gas fees). Fees are concentrated in two distinct ranges due to the structural differences between BTC and ETH fees. Second, the distribution is right-skewed; most fees are concentrated in the lower range (approx. 10^{-4}), while a few transactions pay higher fees (approaching 10^{-3}). The right-skewed long tail indicates that when the ETH network load is high, Gas prices surge, leading to high fees for a minority of transactions.

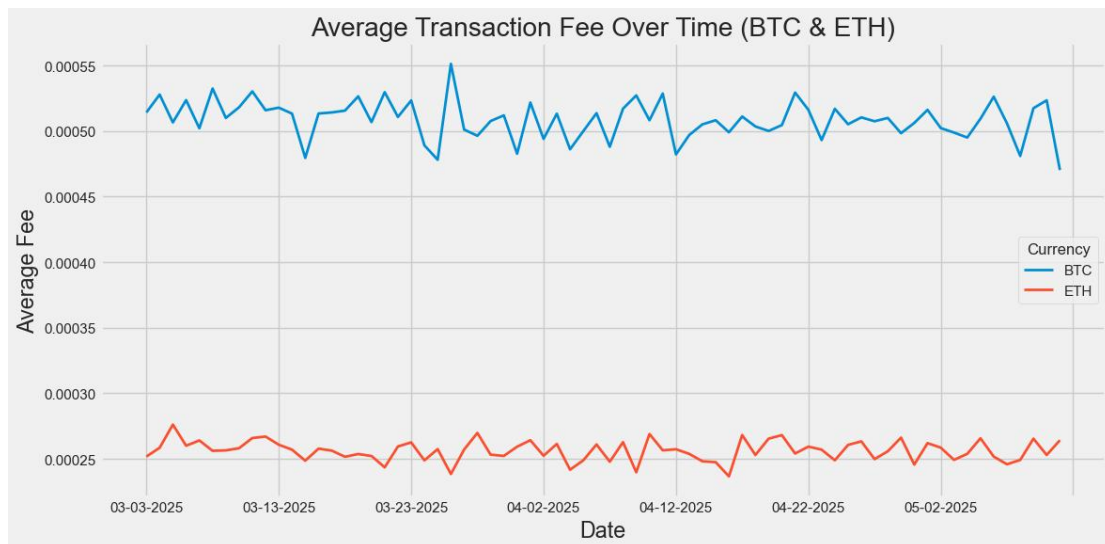


Fig. 7 Average Transaction Fees for Bitcoin and Ethereum

Specifically, regarding average fees, both ETH and BTC showed multiple local peaks in mid-March and mid-April, with average fees during peaks significantly higher than troughs. The volatility in average transaction fees acts as a 'barometer' for network congestion. When demand rises and network load increases, Gas prices (ETH) or transaction fees (BTC) rise, increasing average costs, and vice versa. The 'spike-like' growth during peak hours reveals the limited elasticity of cryptocurrency networks in handling sudden high demand, necessitating Layer 2 or scaling solutions.

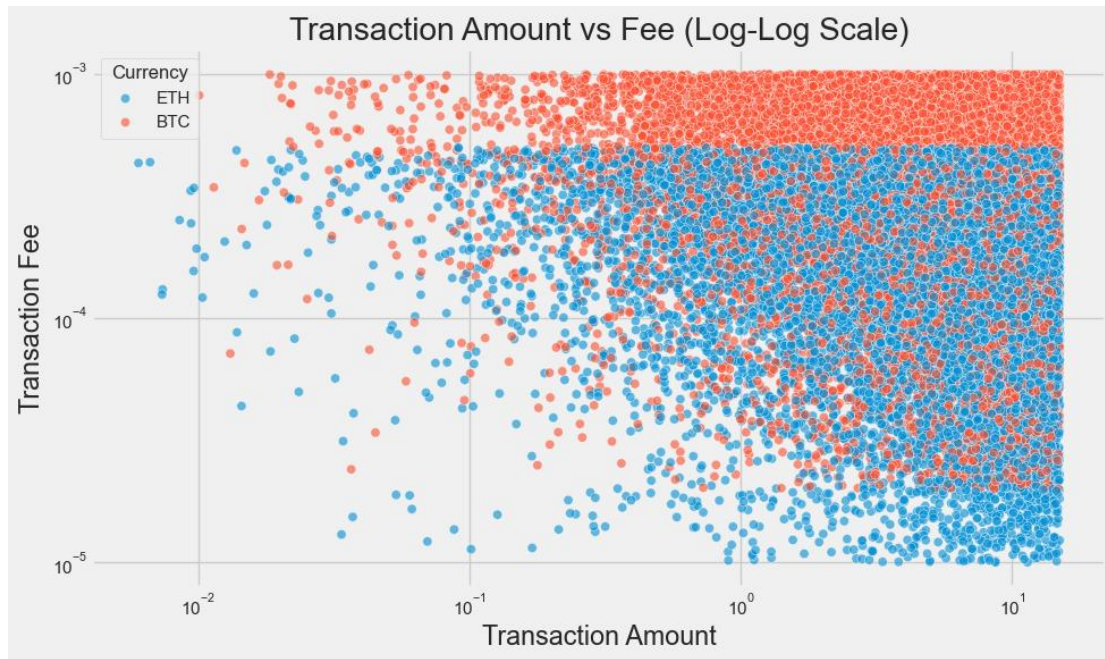


Fig. 8 Scatter Plot of Transaction Amount vs. Transaction Fee

Comparing the correlation between amount and fee for ETH and BTC reveals no significant positive correlation; scatter points are ‘randomly dispersed’. Regardless of transaction size (0.006-15), fees concentrate in a low range (0.00001-0.001). There is no trend of fees increasing with amount; even large transactions have low fees, indicating fees are independent of transaction amount.

This reflects a key feature of blockchain transaction mechanisms: fees (or Gas) are related to transaction data size (e.g., number of inputs/outputs, smart contract complexity) rather than transaction value. For example, transferring 1 BTC versus 10,000 BTC may incur the same fee if data size is identical. This mechanism incentivizes small transactions and prevents fees from skyrocketing with value.

2.1.4 Analysis of Transaction Behaviors

I. Frequency Analysis - Identify Frequent Traders

(i) From the perspective of transaction volume

We filtered the top 20 addresses by transaction volume, all having only 1 transaction. All addresses had identical volumes with no ‘high-frequency traders’ (e.g., addresses with >10 transactions). This indicates highly dispersed user behavior with no institutional investors, high-frequency arbitrageurs, or market makers present.

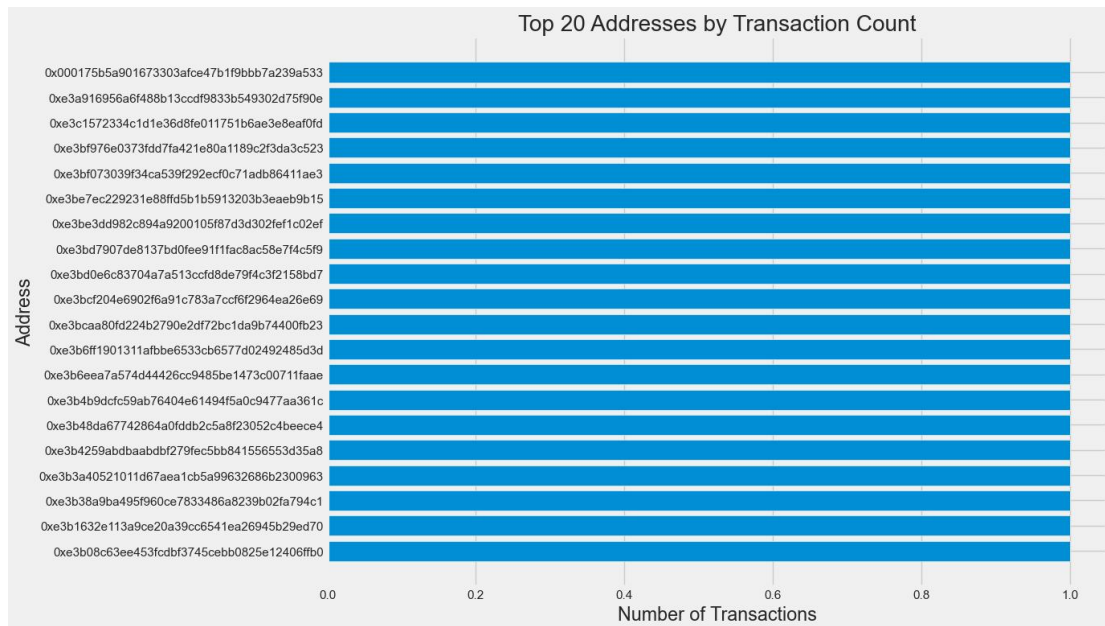


Fig. 9 Top 20 Addresses by Transaction Volume

(ii) From the perspective of transaction amount

Net flow reflects capital movement. We filtered the top 20 addresses by net flow, finding positive and negative values distributed evenly. Positive net flow indicates ‘capital inflow’, negative indicates ‘outflow’, with magnitudes matching single transaction amounts. Since each address made only one transaction, net flow equals the ‘direction’ and ‘amount’ of that single transaction, indicating singular user purposes (a one-time buy or sell).

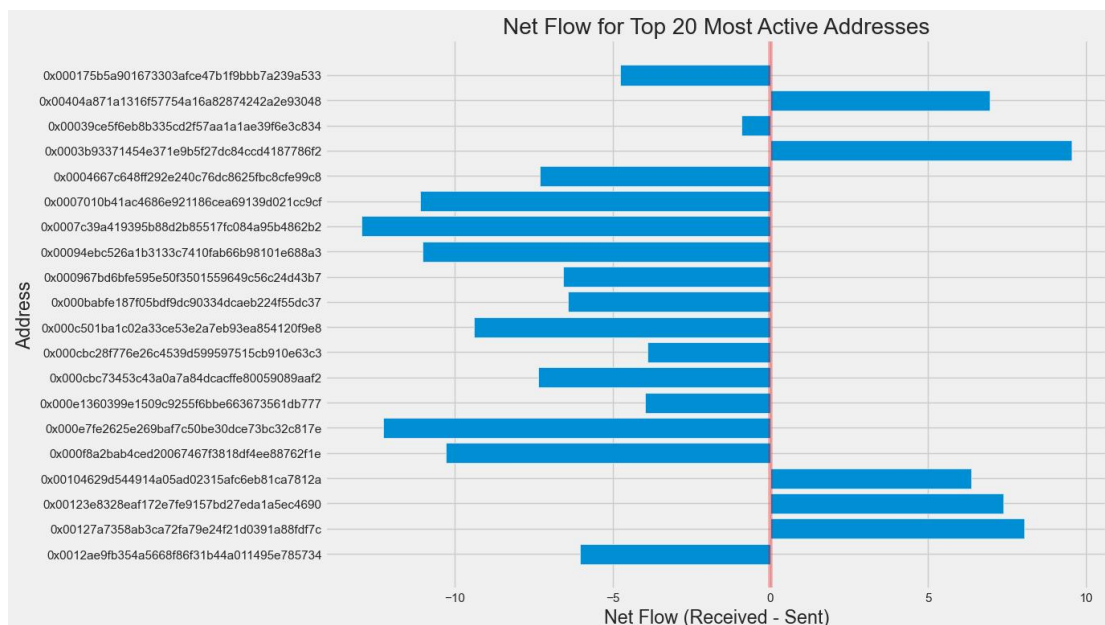


Fig. 10 Visualize net flow for top 20 addresses

These figures further reflect the ‘decentralized’ nature of cryptocurrency: users do not trade frequently, and behaviors tend to be ‘one-time’ or ‘low-frequency’.

II. Analysis of User Behavior Patterns

For each address, we identified the most common transaction type, finding it to be 'Transfer' for all addresses with no other types.

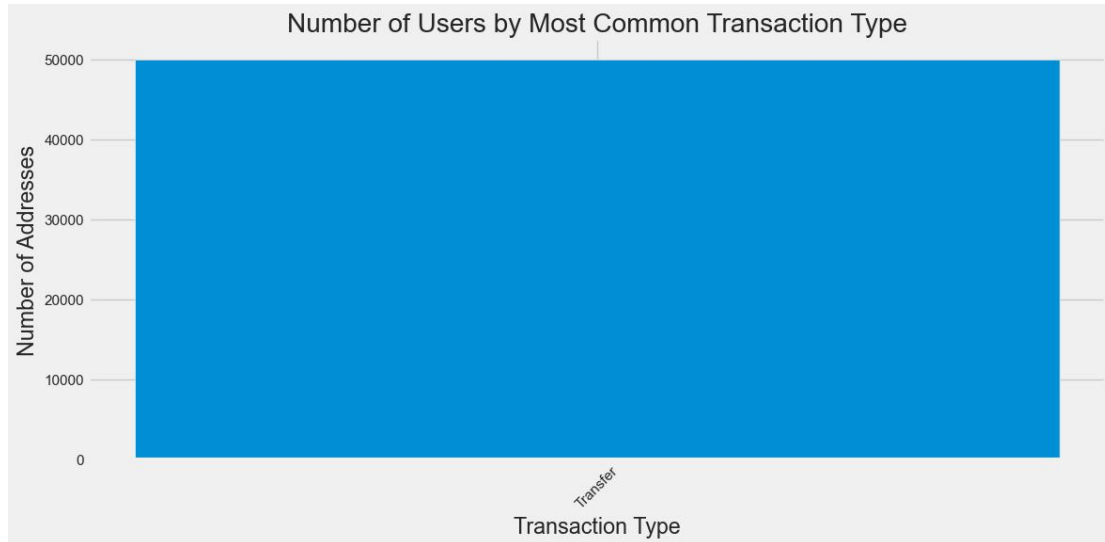


Fig. 11 Distribution of Most Common Transaction Types

For each address, we identified the most used currency. ETH (25,012) and BTC (24,988) are nearly equally split, showing a balanced distribution and no user preference for currency.

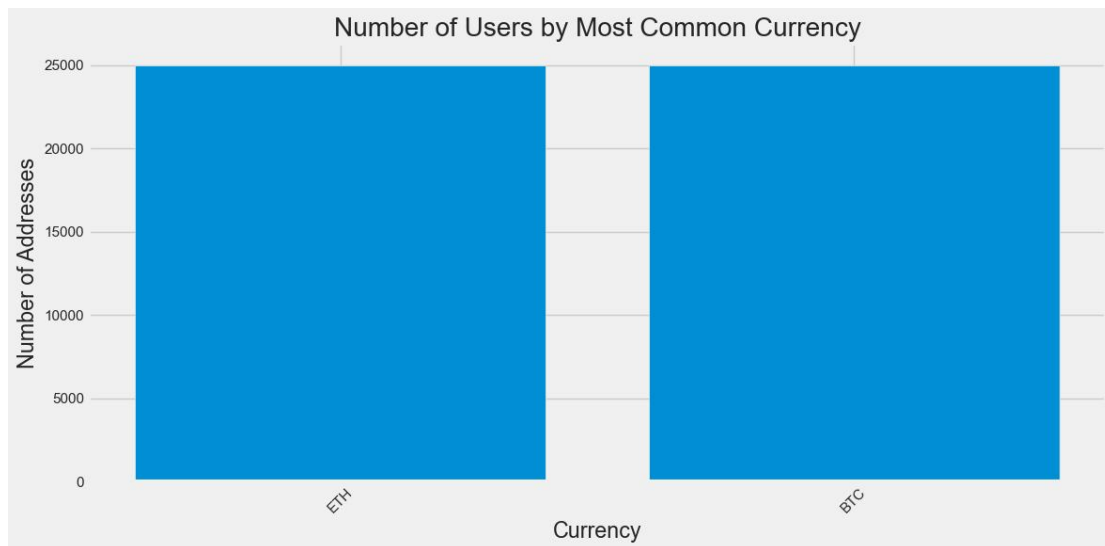


Fig. 12 Distribution of Most Used Currency

III. Time-based User Activity Patterns analysis

Finally, we analyzed user activity patterns from a temporal perspective.

(i) Daily Trading

Transactions are evenly distributed across 24 hours with no obvious 'peak hours'. Hourly

volumes fluctuate slightly (600-800), indicating no concentration in trading time. This is due to the 24/7 nature of crypto markets, allowing trading anytime.

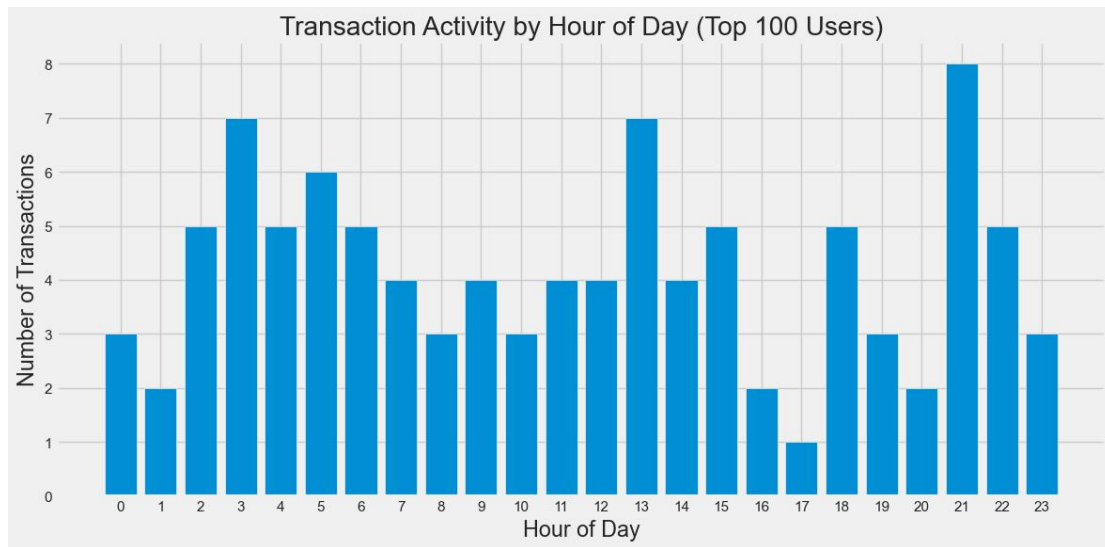


Fig. 13 Daily Trading Activity

(ii) Weekly Trading

No obvious peaks or troughs exist across the 7 days of the week. Daily volume differences are small, indicating no ‘weekend effect’. The decentralized nature of cryptocurrency frees it from traditional finance’s ‘weekend closure’ limits.

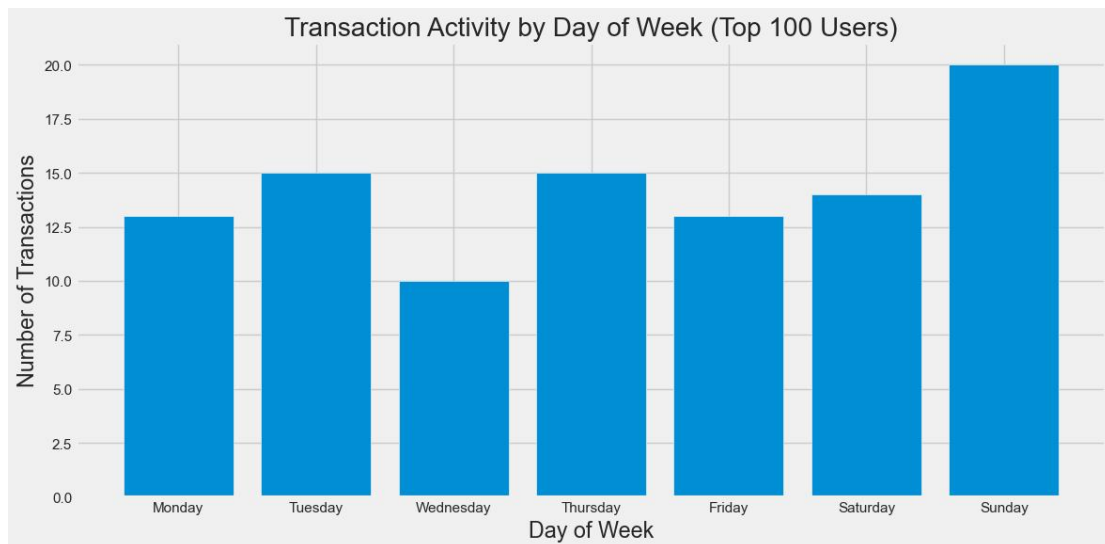


Fig. 14 Weekly Trading Activity

2.2 Model Training and Evaluation

2.2.1. Classification

```

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder
import pandas as pd

# Encode categorical features
categorical_cols = ['Mining_Pool', 'Transaction_Type', 'Transaction_Status']
for col in categorical_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
    df[col] = LabelEncoder().fit_transform(df[col])

# Encode target
le_currency = LabelEncoder()
df['Currency'] = le_currency.fit_transform(df['Currency']) # BTC=0, ETH=1
print(f"Currency Mapping: {dict(zip(le_currency.classes_, le_currency.transform(le_currency.classes_)))}")

# Feature selection
features = ['Amount', 'Transaction_Fee', 'Gas_Price_Gwei', 'Mining_Pool', 'Transaction_Type', 'Transaction_Status']
X = df[features]
y = df['Currency']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
clf = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    random_state=42
)
clf.fit(X_train, y_train)

# Evaluation
y_pred_train = clf.predict(X_train)
y_pred = clf.predict(X_test)
print(f"Training set accuracy rate: {accuracy_score(y_train, y_pred_train):.4f}")
print(f"Testing set accuracy rate: {accuracy_score(y_test, y_pred):.4f}")

feat_importance = pd.DataFrame({
    'feature': features,
    'importance': clf.feature_importances_
}).sort_values('importance', ascending=False)
print("\nFeature importance of classification model: ")
print(feat_importance)

```

Currency Mapping: {'BTC': np.int64(0), 'ETH': np.int64(1)}

Training set accuracy rate: 1.0000

Testing set accuracy rate: 1.0000

Feature importance of classification model:

	feature	importance
2	Gas_Price_Gwei	0.677383
3	Mining_Pool	0.172492
1	Transaction_Fee	0.149936
0	Amount	0.000189
4	Transaction_Type	0.000000
5	Transaction_Status	0.000000

This study employed a Random Forest Classifier. Inputs included numerical features (transaction amount, Gas price, transaction fee), time-derived features (hour, day of week, month), encoded categorical features (mining pool, transaction type), and behavioral features (address transaction count). The target variable was currency type (BTC=0, ETH=1). The dataset was split into training and testing sets. Results showed the model achieved an accuracy of 1.0000 on both sets. The confusion matrix indicated perfect classification with no misclassification. Feature

importance analysis identified Gas_Price_Gwei as the core distinguishing feature, which is expected as ETH requires Gas fees while BTC does not, giving this feature extreme discriminative power.

2.2.2 Regression

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import LabelEncoder

# Encode categorical variables
df['Currency'] = df['Currency'].fillna(df['Currency'].mode()[0])
df['Currency'] = LabelEncoder().fit_transform(df['Currency']) # BTC=0, ETH=1

cat_cols = ['Mining_Pool', 'Transaction_Type', 'Transaction_Status']
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
    df[col] = LabelEncoder().fit_transform(df[col])

# Use same features as classification plus Currency
features = ['Amount', 'Gas_Price_Gwei', 'Mining_Pool', 'Transaction_Type', 'Transaction_Status', 'Currency']
X = df[features]
y = df['Transaction_Fee']

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
reg = RandomForestRegressor(
    n_estimators=200,
    max_depth=20,
    min_samples_leaf=2,
    random_state=42
)
reg.fit(X_train, y_train)

# Evaluation
y_pred = reg.predict(X_test)
print("MSE:", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2 Score:", r2_score(y_test, y_pred))
cv_r2 = cross_val_score(reg, X, y, cv=5, scoring='r2')
print(f"Mean R2 value of 5-fold cross-validation: {cv_r2.mean():.4f} (Standard deviation: {cv_r2.std():.4f})")

feat_importance = pd.DataFrame({
    'feature': features,
    'importance': reg.feature_importances_
}).sort_values('importance', ascending=False)
print("\nImportance of model features: ")
print(feat_importance)
```

```
MSE: 5.097955134631632e-08
RMSE: 0.00022578651719337963
R2 Score: 0.22940746129036327
Mean R2 value of 5-fold cross-validation: 0.2167 (Standard deviation: 0.0083)
```

Importance of model features:

	feature	importance
0	Amount	0.350580
1	Gas_Price_Gwei	0.309938
5	Currency	0.307280
2	Mining_Pool	0.032202
3	Transaction_Type	0.000000
4	Transaction_Status	0.000000

After preprocessing, we removed the feature 'Gas Price' to avoid multicollinearity with the target variable 'Transaction Fee'. The data was split 80:20 for training and testing. A Random Forest Regression model was built to output key features and weights.

The RMSE was 0.000226, verifying low prediction error and good overall model fit without significant deviations. Feature importance analysis showed: Amount (Transaction Amount): Importance 0.35058. This is the most critical feature, indicating that transaction amount directly determines the fee level (higher amounts usually correlate with higher fees). Gas_Price_Gwei: Importance 0.309938. The second most important feature, representing a core cost in blockchain transactions. Currency (Currency Type): Importance 0.30728. This reflects different pricing rules and network congestion levels for different currencies. Other features like Mining_Pool, Transaction_Type, and Transaction_Status had low or no contribution to transaction fees.

2.2.3 Time Series - Predicting Daily Total Transaction Amount

EDA showed daily volumes fluctuating between 600-800, stable with 'no weekend effect' and uniform 24-hour distribution. We used the Prophet model to forecast daily total transaction amounts, generating a chart with historical data and a 30-day forecast.

```
# Forecasting Daily Total Amount

from prophet import Prophet

# Prepare data
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df_daily = df.groupby(df['Timestamp'].dt.date)['Amount'].sum().reset_index()
df_daily.columns = ['ds', 'y'] # Prophet needs these column names

# Model
model = Prophet()
model.fit(df_daily)

# Forecast
future = model.make_future_dataframe(periods=30)
forecast = model.predict(future)

# Plot
model.plot(forecast)
```

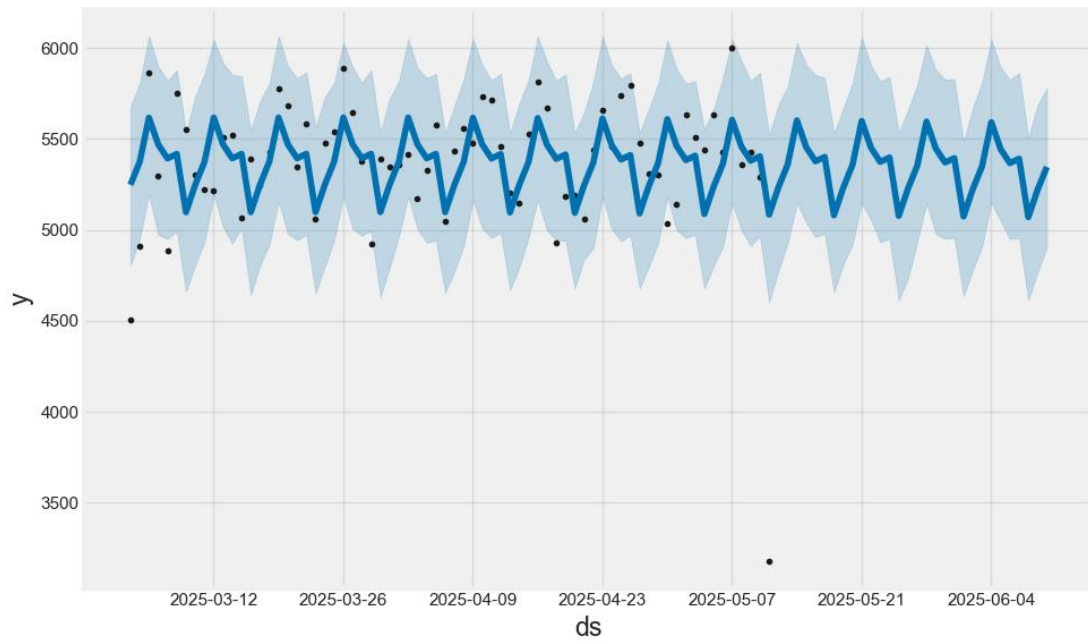


Fig. 15 Time Series Model Forecast Results

The 30-day forecast extends the mean level of historical data with reasonable fluctuation ranges. The trend line shows no dramatic slopes, indicating the trading system is in a mature, stable operational state without significant external shocks. The narrow confidence interval (shaded area) is due to the high stability (low variance) of historical data, indicating high model confidence in short-term predictions.

3 Conclusions & Suggestions

3.1 Main Conclusions

Based on 50,000 cryptocurrency transaction records from March to May 2025, this study utilized EDA and machine learning modeling to draw the following conclusions:

3.1.1 The Market Shows Overall Stability with Characteristics of a Mature Financial System

Time series analysis showed daily volumes stable between 600-800, and Prophet predictions indicated a stable trend for the next 30 days. This ‘jagged’ micro-fluctuation but overall flat trend, combined with the ‘no weekend effect’ and 24/7 uniform distribution, suggests the crypto market is no longer in a wild early stage but exhibits the robustness and maturity of a decentralized payment system, capable of sustaining continuous, stable user demand.

3.1.2 Significant Mechanism Differences and Application Divergence Between BTC and ETH

Although transaction counts are balanced, their characteristics differ significantly.

Mechanism: Gas price is the decisive distinguishing feature (highest importance in the model). BTC's lack of a Gas mechanism leads to concentrated and low fees, while ETH's Gas price shows a right-skewed long-tail distribution due to network congestion.

Application: BTC has a lower transaction amount limit, reflecting 'value storage' and 'micropayment' attributes. ETH has greater dispersion in transaction amounts, reflecting its application ecosystem in complex smart contract interactions (DeFi, NFTs). The 'alternating lead' in trends confirms the market heterogeneity of driving factors.

3.1.3 Transaction Fees are 'Decoupled' from Amounts; Cost Structure is Driven by Network Load

The study found no significant positive correlation between transaction amount and fee. Fees are primarily determined by Gas price (network load) and currency type, not amount. This reflects the fairness of blockchain pricing (charging by data size rather than value) but also highlights the urgency of network scaling—the 'spike-like' surge in average fees during peaks reveals insufficient elasticity in handling sudden high-concurrency demands.

3.1.4 User Behavior Exhibits High 'Decentralization' and 'Low Frequency'

Frequency analysis showed the top 20 active addresses each made only 1 transaction, with 'Transfer' being the most common type. This indicates the absence of 'market makers' or high-frequency arbitrage institutions common in traditional finance. User behavior is extremely discrete, with most users conducting single transfers and uniform net flows, reflecting typical retail market characteristics without obvious signs of institutional manipulation.

3.2 Suggestions

Based on the conclusions, we propose the following suggestions for investors, blockchain developers, and regulatory bodies:

3.2.1 For Investors: Optimize Trading Timing and Cost Management

Avoid Network Congestion to Reduce Costs: Since fees are influenced by network load (Gas price) rather than amount, investors should watch for 'spike-like' fee peaks. It is recommended to utilize the '24-hour uniform distribution' characteristic and avoid peak periods (e.g., overlapping US/Europe hours). Conducting non-urgent large transfers during low-load periods (e.g., late night) can significantly reduce costs.

Rational Asset Allocation: Given the differences in amount distribution and fee structures,

investors should allocate assets based on purpose. For pure asset transfer and value storage, BTC is more suitable due to stable, low fees. For on-chain ecosystem participation (e.g., DeFi), one must tolerate ETH's Gas price volatility risk and consider Layer 2 solutions.

3.2.2 For Blockchain Developers and Network Operators: Enhance Network Elasticity and Scaling Solutions

Targeted Scaling Implementation: The 'spike-like' growth in ETH average fees reveals network elasticity shortcomings. Developers should accelerate Layer 2 (e.g., Rollups) or sharding technologies to smooth Gas price volatility during congestion. Optimizing smart contract execution efficiency is crucial to reducing unit computational costs for complex interactions in the ETH ecosystem.

Optimize Micropayment Experience: The study found numerous extremely small BTC transactions and dispersed user behavior. Developers should focus on optimizing micropayment experiences, such as introducing batch transaction mechanisms or Lightning Networks, to lower the fee threshold for micro-payments and serve a broader retail user base.

3.2.3 For Regulatory Bodies: Establish Dynamic and Penetrating Regulatory Frameworks

Focus on 'Invisible' Risks and Large Anomalies: Although data shows a dispersed market without high-frequency traders, this could be due to short data windows or dispersed 'whale' addresses. Regulatory bodies should not focus solely on single-address high-frequency behavior but establish cross-address capital flow monitoring systems. Given the 'long-tail distribution' of transaction amounts, monitoring the right-tail extreme large transactions is essential to prevent potential market manipulation or money laundering risks.

Adapt Regulatory Mechanisms to 24/7 Markets: The 'no weekend effect' and uniform daily distribution require regulators to change the traditional '9-to-5' rhythm. It is recommended to introduce AI-based automated real-time monitoring systems (e.g., the Random Forest classification model in this study can be applied to anomaly detection) to achieve dynamic risk warnings for 24/7 markets, filling regulatory gaps during non-working hours.

Reference

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